Computational geometry solutions

A10-P01 Orientation and sign translation

b−a=(4,1) and c−a=(2,3), giving orient(a,b,c)=4 · 3−1 · 2=10. Reversing the vectors gives 2 · 1−3 · 4=−10. Since q−a=(8,2), orient(a,b,q)=4 · 2−1 · 8=0. Triangle area is |10|/2=5.

CLRS DIRECTION reverses the operands, giving −10,10,0; a left turn requires a negative value. If mathematical y is drawn downward without conversion, the figure is reflected vertically and a positive turn looks clockwise. Convert drawing coordinates or explicitly redefine the convention; do not mix them.

A10-P02 Boundary cases are part of the specification

Case	(o1,o2,o3,o4)	Closed	Proper	Reason
I	(0,0,0,0)	Yes	No	on(a,b,c); overlap x∈[2,6]
II	(−6,12,21,3)	No	No	Second sign pair agrees; all on tests fail
III	(0,0,0,0)	Yes	No	on(a,b,c); intersection is (6,0)
IV	(−12,12,12,−12)	Yes	Yes	Both sign pairs oppose; crossing at (3,0)

The distinct point segments also produce four zero orientations, but all on tests fail. A zero vector has zero cross product with every vector; this does not make two points equal. Both coordinate bounds remain necessary.

A10-P03 Arithmetic and comparators

Each difference is at most 2B in magnitude. Each product is at most $4B^2$, and their difference is at most $8B^2$. At B=100000 this bound is 80000000000, exceeding signed 32-bit maximum 2147483647. The actual points (0,0),(100000,0),(0,100000) produce 10000000000. A bound exceeding the type means no guarantee; this instance proves overflow can occur.

Subtraction itself can overflow: 2147483647−(−2147483648)=4294967295. Casting an already corrupted result cannot recover it. Widen both operands first. Even signed 64-bit determinants are not guaranteed for full signed 32-bit coordinates.

Multiplying coordinates by λ multiplies orientation by λ^2. A fixed ε can classify the same shape differently at different scales. The values 0 and 0.75ε are close, as are 0.75ε and 1.5ε, but 0 and 1.5ε are not. Approximate equality is not transitive, so it cannot directly serve as a tree or sort equivalence relation.

A10-P04 Deletion exposes an intersection

x	Event	Bottom-to-top status	Tests
0	Insert a	[a]	None
1	Insert c	[a,c]	a,c disjoint
2	Insert b	[a,c,b]	c,b disjoint
3	Before deleting c	[a,c,b]	a,b intersect; stop

The equations are y=x/2 for a and y=(54−6x)/7 for b. Equality gives 7x=108−12x, hence x=108/19 and y=54/19. This x belongs to a's [0,10] and b's [2,9]. The supporting line of c meets a at x=4 and b at x=20/3, both outside c's [1,3].

Insertion-only tests miss a,b because c separates them at their insertion events, with no further insertion before the crossing. A complete-segment predicate already proves a future meeting, so detection can return at x=3. Status maintenance ends there, and no intersection swap event is needed.

A10-P05 Tied events, verticals, and shared points

Insert b before deleting a at x=2. They coexist in status, allowing their shared endpoint (2,1) to be detected. Reverse ordering misses it. Interval [3,4] has no hit among 1,2,6. Interval [2,4] hits y=2; strict inequalities would miss that boundary.

A generalized x batch inserts nonvertical starts, queries vertical segments, then deletes nonvertical ends. Checks against the nonvertical tree do not detect vertical-vertical overlap. Check those closed intervals separately at the same x. Point intervals cover zero-length segments.

An id only makes ordering reproducible. It does not check equal-height groups or establish the neighbor argument. Triple intersections require explicit grouped-event rules and an updated proof; otherwise retain the simplifying assumption. The height-query example is independent of the shared-endpoint example and must not be conflated with its coordinates.

A10-P06 Detection is not reporting

At x=1 the heights are a=1,b=3,c=5, so status is [a,b,c]. At x=3 they are a=3,b=1,c=5, giving [b,a,c]. The crossing (2,2) is not an endpoint event. Printing without swapping leaves the tree in [a,b,c], which no longer represents current geometric order. This proves an invariant failure; it does not claim that these three segments necessarily cause another pair to be missed.

Reporting dynamically schedules future intersections of neighbors, exchanges or reorders status at crossings, updates neighbor events, suppresses duplicates, and rejects stale events. Its general-position time is $O((n+k)\log n)$. Degenerate inputs also require output rules for multiple pairs at one point, verticals, and collinear overlap.

Take horizontal segments $(0,i)\to(m+1,i)$ for $i=1,\ldots,m$, and vertical segments $(j,0)\to(j,m+1)$ for $j=1,\ldots,m$. Each horizontal meets every vertical once. Thus $n=2m$ and $k=m^2=n^2/4$ distinct intersections. Explicitly outputting all of them requires $\Omega(k)$, ruling out an $O(n \log n)$ worst-case reporting bound.

A10-H01 Polar-angle ties

Raw I,B,C,D,H,E,J,F,G; remove I,H,J; keep B,C,D,E,F,G. Each removed point is a convex combination of A and the farthest point on its ray.

A10-H02 Replay the stack

B→A,B; C→A,B,C; D pops C→A,B,D; E→A,B,D,E; F pops E→A,B,D,F; G→A,B,D,F,G. orient(B,C,D)=−3 and orient(D,E,F)=−4.

A10-H03 Collinear output

Return endpoints (0,0),(2,0) after coordinate deduplication. Changing $\leq$ to $<$ alone is insufficient. Angle-tie removal, all-collinear ordering, and duplicate endpoints also need a consistent policy.

A10-H04 Choose a hull method

Graham is $O(n \log n)$ in both cases. Jarvis is $O(n)$ when $h=4$ and $O(n^2)$ when $h=n$. h counts output extreme vertices, not interior points.

A10-H05 Split repeated x values

X=A,B,C,D,E,F,G,H; L=A,B,C,D; R=E,F,G,H. YL=A,C,B,D; YR=G,E,F,H. D and E share x=0 but belong to different sides. Coordinate-only assignment can unbalance the split and fails to shrink an all-covertical input.

A10-H06 Compute the strip

Left 10, right 6.25, $\delta=2.5$. Strip C,E,F,D. Candidate squared distances are C-E=1.25, C-F=13, E-F=6.25, F-D=8. The answer is C-E with $\sqrt{1.25}\approx1.118033989$. An updated bound can skip later candidates.

A10-H07 Prove the successor bound

Four $\delta/2$ cells per side have diagonal $\delta/\sqrt{2}<\delta$. Same-side separation $\geq\delta$ allows at most one point in each cell. There are at most eight points including p, hence seven later points. Line points retain their assigned side. Return immediately when $\delta=0$.

A10-H08 Repair the complexity and duplicates

$T(n)=2T(n/2)+O(n \log n)=O(n \log^2 n)$. Removing equal coordinates can lose a zero-distance pair of distinct ids. Detect such a pair first using adjacent equal coordinates in a coordinate sort, or keep all ids through recursion.

References and attribution

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https://github.com/xinhi6716/computer-science-self-study